

LAB 3

Component Modelling & Architectural Pattern Selection

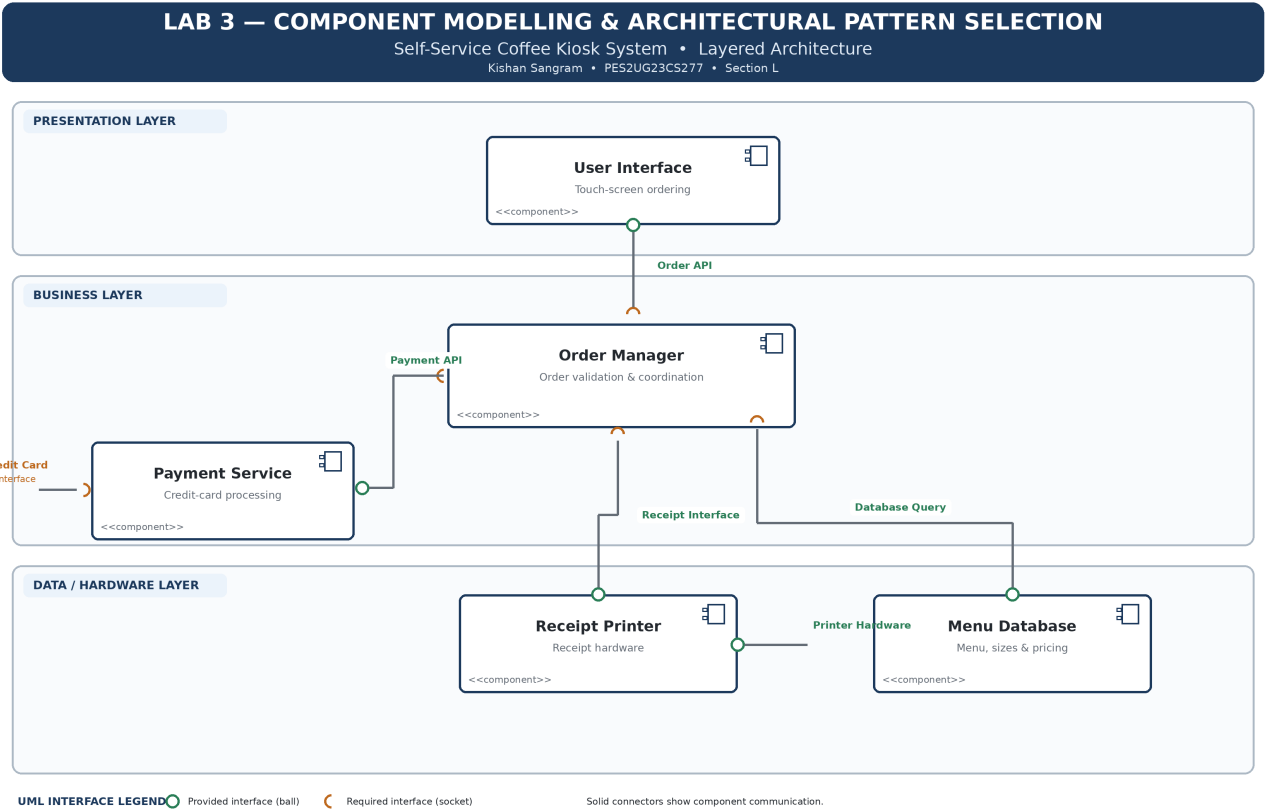
Self-Service Coffee Kiosk System

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Architecture Selected: Layered Architecture

The kiosk uses a presentation layer for touch-screen interaction, a business layer for order and payment coordination, and a data/hardware layer for menu data and receipt output.

Component Diagram



Written Justification

Choice: Layered Architecture. **Reasons:** clear separation of UI, business, payment, printing and data responsibilities; and a direct fit for a small traditional kiosk without microservice operational complexity. **Security:** payment handling is isolated behind a defined Payment API. **Performance:** local communication avoids unnecessary network hops and latency.

Architecture Analysis & Component Identification

Layered: Clear separation of concerns and easy understanding; limitations include layer coupling and performance overhead.

Microservices: Independent scaling, fault isolation and deployment; limitations include operational complexity, network latency and data consistency challenges.

Client-Server: Centralized control and simple deployment; limitations include a single point of failure, scalability bottleneck and limited fault tolerance.

Components

1. User Interface Component
2. Order Manager Component
3. Payment Service Component
4. Receipt Printer Component
5. Menu Database Component

Interfaces

- Order API — User Interface ↔ Order Manager
- Payment API — Order Manager ↔ Payment Service
- Receipt Interface — Order Manager ↔ Receipt Printer
- Database Query — Order Manager ↔ Menu Database
- Credit Card Interface — Payment Service ↔ external card-processing endpoint
- Printer Hardware — Receipt Printer ↔ physical printer

Final checklist: 5 components ✓ 4+ interfaces ✓ Ball/socket notation ✓ Data-flow labels ✓ Security ✓ Performance ✓